

An improved cell situation identification approach with convolutional neural network and wavelet extreme learning machine

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Abstract

In the aluminum reduction process, the flame hole is an influential index of the whole production situation, which reflects the distribution of the physical field of the reduction cell, the current efficiency and the lifespan of the aluminum reduction cell. Therefore, flame hole situation detection and identification are critical and significant in the whole process of aluminum electrolysis. However, in the practical industrial production, flame hole identification result is coming from the experimental operation workers, with the loss of workers and different experimental levels, the real-time measurement of the flame hole index is still a challenge beyond solution. This article develops a flame hole image classification method based on wavelet extreme learning machine. First, a deep feature set is extracted from the original images with a convolutional neural network. Then, a classification model based on an extreme learning machine with wavelet activation function is developed. Finally, the proposed convolutional neural network–weighted extreme learning machine model is applied to superheat degree real-time detection in the industrial electrolysis cell. The proposed method is evaluated on aluminum production which outperforms existing other superheat degree methods in accuracy and robustness.

Keywords

Flame hole images, extreme learning machine, aluminum electrolysis, cell classification, wavelet neural network, convolutional neural network

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Introduction

In the aluminum electrolysis production process, superheat degree (SD) is a critical indicator which reflects the current efficiency and quality of aluminum production. In the actual production process, the traditional method is to use the manual experience to identify the state of the flame hole. However, the electrolytic aluminum industry is a high-temperature, high-pressure industry, containing a variety of harmful gases in the hard environment when workers work in the factory for a long time, and it costs a lot for detecting key SD index. With the loss of experimental workers and the error of manual experience, the real-time measurement of the flame hole becomes a big challenge in the production processes.

Flame hole images are defined as the outline of the electrolyte, which contains many information about the production. Color change feature is commonly extracted according to the variation of the brightness

value of the fire area of the sampling frame. Compared with these hard-to-measure methods, the judgment conduct by experienced workers according to the tank condition cannot guarantee the accuracy. Due to the limitations of experienced operators, the judgment has many shortages, such as space, time, and subjectivity. Recently, with the development of machine learning, some improved and efficient algorithms such as radial basis network (RBN), support vector machine (SVM), and convolutional neural network (CNN) have shown great performance in some industrial domains, which gives us a new insight to build the proposed method to detect the cell situation.

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In the last decades, neural network has become an efficient image processing method which has been widely used for many classification tasks. For example, Huang and Wang¹ proposed a convolutional neural network to classify Chinese painting images. He et al.² applied SVM to the classification of multi-source remote sensing images. However, with the increment of the feature dimension, the performance accuracy of SVM would be greatly reduced. Furthermore, some deep learning methods have been proposed to detect these processes' variable, while these approaches greatly rely on selecting the manual features. The denoising ability is also not satisfactory in industrial applications. Extreme learning machine (ELM) is a single hidden layer forward neural network algorithm proposed by Huang et al.,³ which has been applied to various fields and achieved good performances. For example, Qzyurt⁴ applied the ELM to the diagnosis of breast cancer disease and improved accuracy. Wu et al.⁵ investigated the ELM in the field of fault diagnosis. Li and Guo⁶ developed a novel ELM method for face recognition. Other ELM applications have been extended to pattern recognition, industrial monitoring, and video processing.

A commonly used hidden activation function in the existing neural networks is Sigmoid function. The projection of the Sigmoid can be given as the following equation

$$g_1(a, b, x) = \tan h(x) = \frac{e^{(a \cdot x + b)} - e^{-(a \cdot x + b)}}{e^{(a \cdot x + b)} + e^{-(a \cdot x + b)}} \quad (1)$$

where a is the input weights and b is the corresponding bias. x is the input label. However, Sigmoid activation function has the difficulty to establish the accurate mapping relationship between the various patterns and the raw input signals.⁷ Moreover, in practical engineering, the processes variables collected from the aluminum production plant are always very complex and non-stationary with heavy noise. Another activation function in the radial basis function (RBF) network is the Gaussian activation, and some recent works about Gaussian have shown good performance.^{8,9} The mathematical equation of Gaussian is

$$g_3(a, b, x) = e^{\left(-\frac{b\|x-a\|^2}{2}\right)} \quad (2)$$

Gaussian kernel models have the difficulty to improve the performance when the distribution of the input patterns is not the Gaussian geometry. The Gaussian function can only extract the time temporal features, and some important frequency characteristics are ignored. The Gaussian's long tail from negative infinity to plus infinity gives it poor denoising capability.

Wavelet transforms contain the scale parameter and shift factor, which has been successfully employed as the new activation functions of shallow neural networks to construct the so-called wavelet neural networks (WNN).^{10–13} The shift factor enables the wavelet to perform ergodicity analysis along the time axis of

the signals. The scale factor aims to shrink and stretch wavelets to approximate the signals with different frequencies for each ergodicity analysis.^{14,15} Therefore, wavelet functions have good time-frequency localization property and focal features through the change in different scale factors and shift factors. Lots of investigations have proved that WNNs usually show obvious advantages over the traditional neural networks in various applications.^{16,17} However, there are a few types of research on wavelet functions applied in industrial flame hole tasks. Thus, it is meaningful to develop novel CNN models that combine the advantages of wavelet functions and deep learning.^{18–20}

Due to the reason that the ELM has the advantages of fast training speed and high classification accuracy and CNN has a good feature representation for these flame hole images,²¹ a novel method of fire hole image classification based on the fusion of CNN and ELM is proposed in this article. The main contributions of this article are as follows:

1. For extracting the features, an improved novel SD identification method with CNN and wavelet ELM is build.
2. The traditional neural network mainly relies on the manual feature selection, and the denoising ability of these methods is limited. The proposed weighted extreme learning machine (WELM) utilizes the wavelet activation function which has significant bandwidth and good denoising performance.
3. The proposed methods coherent the fast training ability of ELM. The training complexity of the proposed method is hundreds of times faster than the backpropagation-based methods.

The industrial cases in aluminum reduction cells evaluated the effectiveness of the proposed method.

The rest of this article is organized as follows. In section "Related works," the related preliminary work about ELM is introduced. We describe the main algorithm framework in section "Methodologies." In section "Experiment results," the experiment results are given and the conclusions are drawn in section "Conclusion."

Related works

In this section, a brief introduction of the processing of ELM is presented.

ELM

ELM is a new training algorithm to solve the single hidden layer feedforward neural network with high accuracy and training speed. Its network structure is shown in Figure 1. Assume that the given training sample set is composed of N different samples, that is, (x_i, y_i) , $i = 1, 2, \dots, N$, where $x_i = [x_{i1}, x_{i2}, \dots, x_{in}] \in R^n$, belong to input variables, $y_i = [y_{i1}, y_{i2}, \dots, y_{im}] \in R^n$, is the

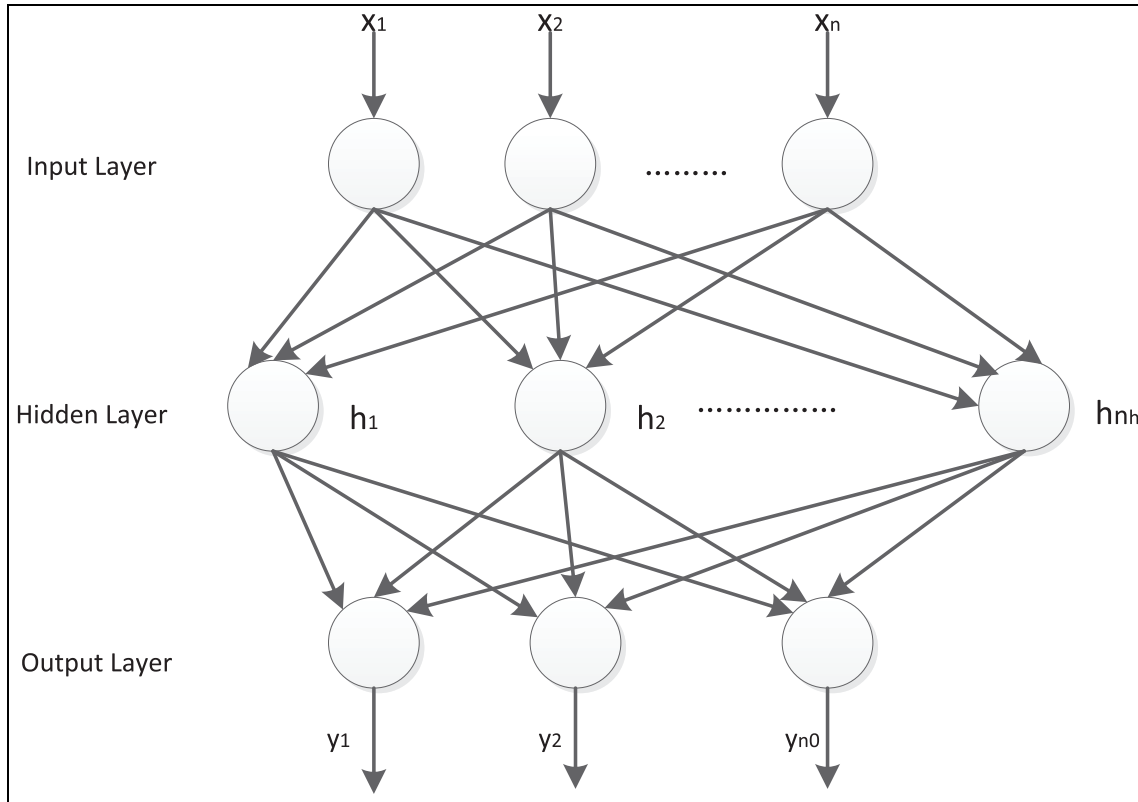


Figure 1. Structure diagram of extreme learning machine.

output class label corresponding to the input variable. The goal of the ELM is to find a suitable nonlinear function to estimate the value of the output Y . Different from the traditional feedforward neural network training method, the training consists of two stages: random mapping and the parameters solving. In the random mapping stage, the hidden layer with a fixed number of randomly generated mapping neurons is constructed, and the mapping is usually expressed by the Sigmoid function or Gaussian function

$$\mathbf{J}(x; \theta) = (1 + \exp(-(w^T x + b)))^{-1} \quad (3)$$

$$\mathbf{J}(x; \theta) = \exp(b\|x - w\|) \quad (4)$$

where $\theta = w, b$ are the parameters of the mapping function, and $\|\cdot\|$ denotes the Euclidean norm.

Before the training of the ELM, the input weights a connect the input layer and the hidden layer, and the neuron threshold b of the hidden layer has been randomly generated. Therefore, in the parameter solving stage, the output weights β between the hidden layer and the output layer can be obtained by solving a regularized least square problem.

We define the output vector of hidden layer as $\mathbf{J}(x_i \in \mathfrak{R}^{N \times n_h})$, where n_h is the number of hidden neurons. The output weight is defined as $\beta \in \mathfrak{R}^{n_h \times n_o}$. Therefore, the output of the network is expressed as follows

$$\hat{y}_i = \mathbf{J}(x_i)\beta \cdot i = 1, 2, \dots, N \quad (5)$$

The output weight between the hidden layer and output layer is obtained by solving the least squares solutions of the following equations

$$\min_{\beta \in \mathfrak{R}^{n_h \times n_o}} \psi_{ELM} = \frac{1}{2} \|\beta\|^2 + \frac{C}{2} \|\xi\|^2 \quad (6)$$

$$\xi = \mathbf{Y} - \mathbf{J}\beta \quad (7)$$

where $\mathbf{J} = [J(x_1)^T, \dots, J(x_N)^T]^T \in \mathfrak{R}^{N \times n_h}$. In equation (4), the first term is the regularization term against over-fitting, and the second is the error vector, which can be presented as equation (5), and C is the penalty coefficient. By setting the gradient of ELM with respect to β to zero, we can get the following equation

$$\nabla \psi_{ELM} = \beta + \mathbf{C}\mathbf{J}^T(\mathbf{Y} - \mathbf{J}\beta) = 0 \quad (8)$$

When the number of the training samples is larger than the number of hidden neurons, which means the hidden matrix $\mathbf{J}$ is full column rank, equation (6) can be solved as follows

$$\beta^* = \left(\mathbf{J}^T \mathbf{J} + \frac{I_{n_h}}{C} \right)^{-1} \mathbf{J}^T \mathbf{Y} \quad (9)$$

where I_{n_h} is the identity matrix of dimension n_h , which is a square matrix with diagonal elements are "1" and others are "0."

Otherwise, when the number of the training samples is less than the number of the hidden neurons, which

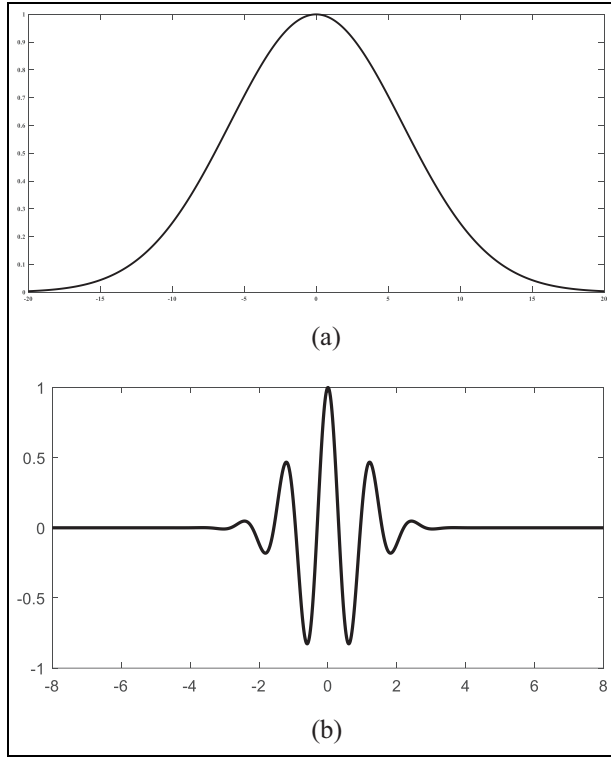


Figure 2. Comparison of the Gaussian kernel and wavelet kernel, and the Morlet wavelet have a better convergence performance: (a) Gaussian kernel and (b) Morlet wavelet activation.

means the hidden matrix $\mathbf{J}$ is full row rank, the solution is given as follows

$$\beta^* = \mathbf{J}^T \left(\mathbf{J}^T \mathbf{J} + \frac{\mathbf{I}_N}{C} \right)^{-1} \mathbf{Y} \quad (10)$$

where $\mathbf{I}_N$ is the identity matrix of dimension N .

The detail of the traditional ELM algorithm procedure is stated in Algorithm 1.

Algorithm 1. ELM algorithm

Input: Training set $\{(x_j, t_j)\}_{j=1}^N$; activation function $J(x)$
Output: The output weight matrix β for ELM; The mapping function of g .
 Step 1: Initiate an ELM framework of n_h hidden neurons with random weights and biases;
 Step 2: Calculate the output of the hidden neurons $\mathbf{J}$
 Step 3: If $n_h \geq N$, compute the output weights using equation (7); Else, compute the output weights using equation (8);
 step 4: Compute the projection function of ELM.
return The output weight matrix β ;

Wavelet transform

The wavelet activation function is proposed by Zhang. Two factors are used in the discrete wavelet transform, which is the scale and transmission factors. Due to the

localization properties in time and scale, wavelet transformation of such signals can allow a machine learning model to better analyze the time evolution of hydrological processes at different scales. Hence, wavelet decomposition of inputs is considered a multi-resolution analysis (MRA) tool for pre-processing non-stationary signals. This technique decomposes the input signals to extract time and frequency components with different shifts and scale information. Furthermore, the Gaussian function has a long tail from the infinity to the plus infinity which restricts the ability to denoising the input signals (Figure 2).

The commonly used activation function in neural networks is discrete wavelet transform (DWT), which is

$$\text{DWT}(a, b) = \frac{1}{\sqrt{a}} \sum_i x(t) \cdot \psi\left(\frac{t - b_i}{a_i}\right) \quad (11)$$

where a and b are its corresponding scale (dilate) and translate (shift) factors. ψ is the wavelet function. Due to the different transmissions, the scale and translation factors from equation (11) are evaluated on a discrete grid of time scale to generate scaled and shifted daughter wavelets from a given mother wavelet.

The DWT can be approximated as a single hidden layer neural network. Indeed, equation (11) can be seen as the output expression of a single hidden layer feed-forward neural network (SLFN) that would have an activation function for a hidden node, with a linear neuron in the output layer. Morlet neural network has been a classic kernel that has applied to many domains.

The real part of Morlet activation function is given as the following equation

$$\varphi(x) = \cos(5x) \exp(-x^2/2) \quad (12)$$

where x is the input of signals.

To get some prior knowledge or training procedure, we transmit the front information to the layer neurons to form our proposed WELM.

Methodologies

In this section, we briefly introduce the image pre-processing and give the structure diagram of the algorithm in this article.

Image pre-processing

The flame hole images used in this experiment were obtained from the aluminum electrolysis plant in Weiqiao, Shandong, China, using an all-digital industrial camera. The data set contains a total of 300 flame hole images of different sizes. According to the images, aluminum electrolysis tanks are divided into three categories: “Hot” label, “Normal” label, and “Cold” label. Among them, there are 79 images of the hot tank, 125 images of the normal tank, and the remaining 96 images represent images of the cold tank.

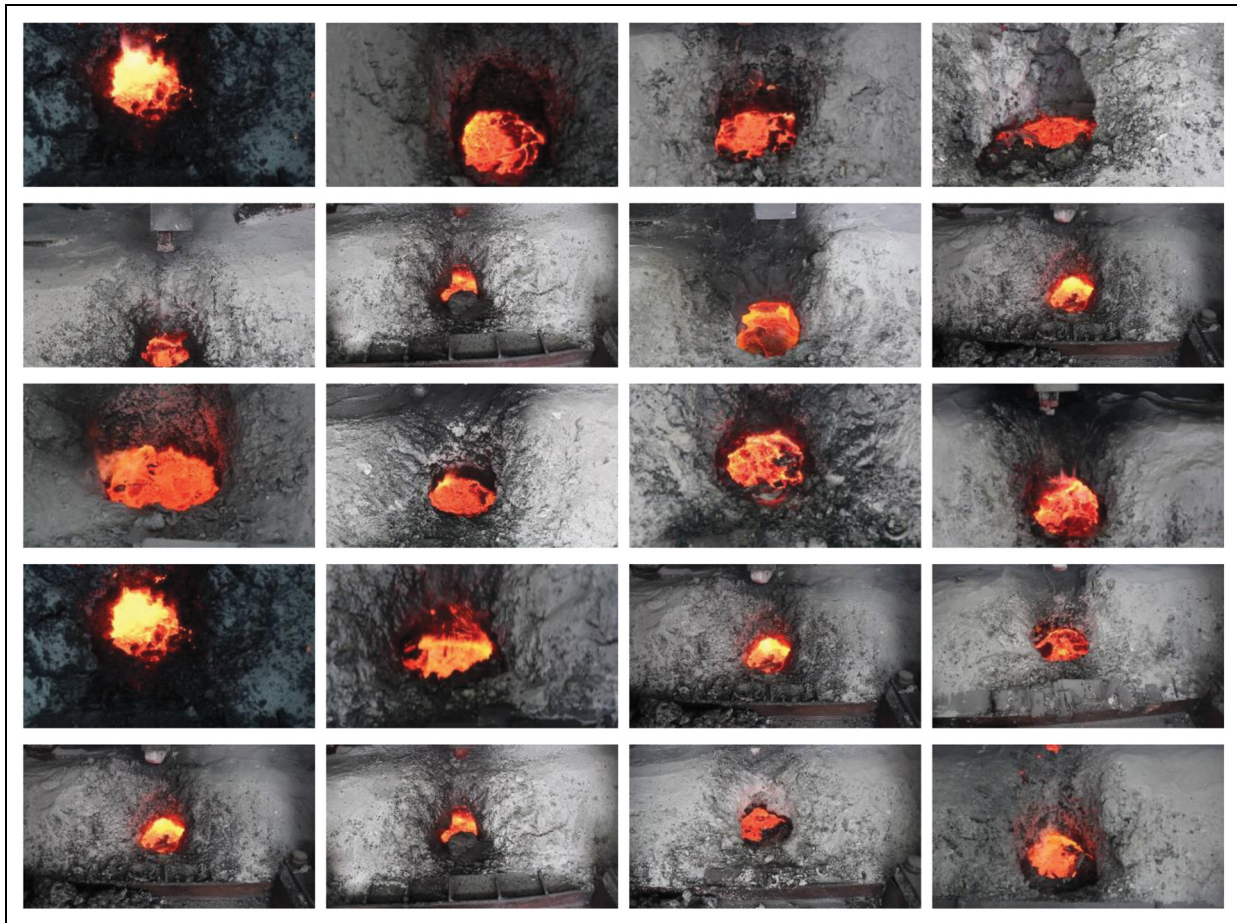


Figure 3. The original flame hole image from the industrial plant.

Normalization. Due to the influence of the environment around the electrolytic cell, the fixed position of industrial digital camera is different in each electrolytic cell, which leads to the relative position and size of the hole part in the acquired images. Before image segmentation, each image is normalized to a size of 1024×1024 dimension, and the normalized image is shown in Figure 3.

Image segmentation. The flame hole image is composed of two parts: the background and the flame hole. If the background feature is wrongly regarded as the part of the flame hole feature, it will directly affect the judgment of the electrolytic cell condition. In order to accurately extract features and avoid interference from the background image to the feature extraction of the flame hole part, it is necessary to distinguish the flame hole from the background. Generally, in the collected images, the proportion of the flame hole in the whole image is relatively small. If the traditional intergroup variance method is used for image segmentation, the flame hole mouth cannot be accurately separated. According to this characteristic of the flame hole image, we select an appropriate threshold value for segmentation by using the red channel according to the RGB distribution of flame hole image. When the pixel value of the image pixel in the red channel is greater

than 230, the region is considered as the flame hole part (Figure 4).

Feature representation with CNN

As in Figure 5, the original images segmented to a dimension 48×48 , a convolutional neural network with a six-layer network which includes two convolution layers, two max-pooling layers, and two fully connected layers. The first Conv layer is a three-channel layer with a $32 \times 3 \times 3$ kernels. Then, a pool operation kernel is multiplied by the Conv layer, and then, an output of the second layer is $32 \times 2 \times 2$. Finally, a 256-dimension deep feature input matrix is extracted from training flame hole images.

ELM for the final classifier

The 256-dimension matrix is an input for the top-layer ELM classifier. Compared with the traditional ELM, the Morlet wavelet kernel is used to the nonlinear activation function.

The improved wavelet ELM approach is presented as follows: The algorithm framework used in this article is shown in Figure 5. The pseudo-code of this method are given in Algorithm 2.

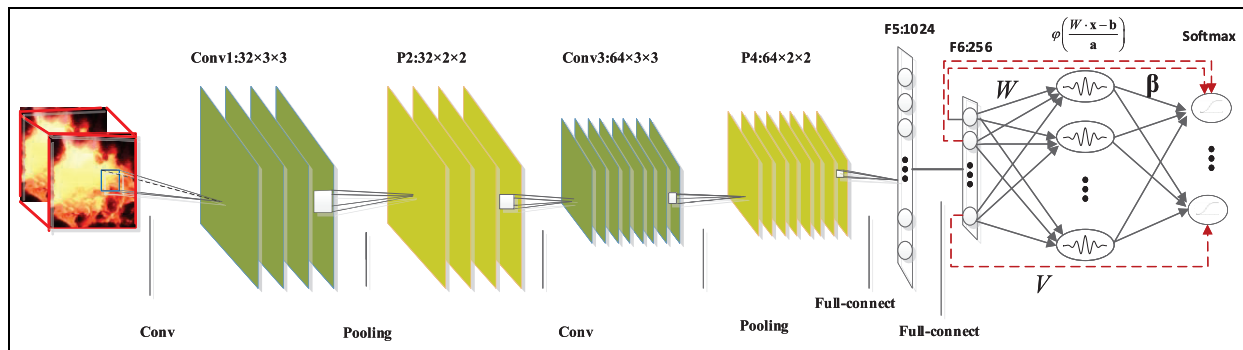


Figure 4. The specification of our proposed method.

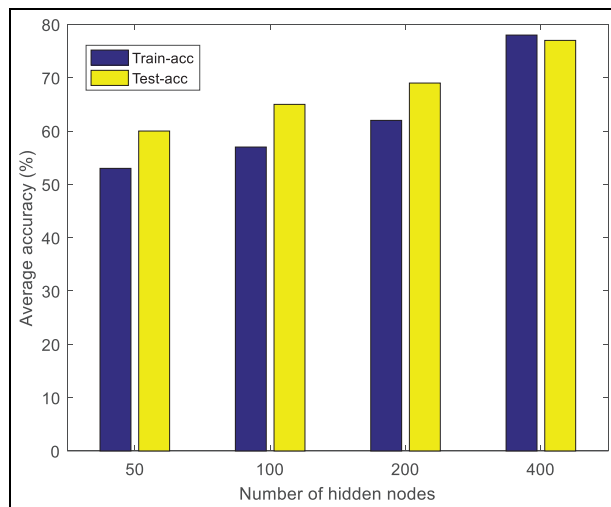


Figure 5. The accuracy of CNN-RBF with the different hidden units. Train-acc and Test-acc denote the training accuracy and testing accuracy, respectively.

Experiment results

To verify the effectiveness of the proposed algorithm, CNN-WELM is applied to the recognition of superheat in industrial aluminum electrolysis process. In this section, the algorithm proposed in this article is compared with other algorithms such as CNN, ELM, and SVM, and the experimental results are given.

The experiment environment is based on MATLAB 2016a running on a 3.6 GHz i7 CPU with 32-GB RAM. Other parameters of CNN include the batch size, learning rate, and activation function. The batch size in our experiment is set to 20, and the learning rate η is equal to 0.003. Tanh and Sigmoid activation functions are also chosen for a comparison. In the Morlet transform, the shift factor was set to 4, and the scale is set to zero. All the experimental results are conducted with three-trial average cross-validation.

In an electrolytic aluminum industry, when the superheat is between 2 °C and 8 °C, the electrolytic cell is considered as a “LOW” cell; when the superheat is between 8 °C and 15 °C, the electrolytic cell is considered as a “NORMAL” cell; and when the superheat

Table 1. Experiment results with different methods.

Methods	Accuracy	Training time	RMSE
SVM	0.54	2.14	0.2234
ELM	0.62	3.36	0.5673
WELM	0.78	2.14	0.5666
CNN-ELM	0.82	1.83	0.8374
CNN-WELM	0.94	1.69	0.0459

SVM: support vector machine; ELM: extreme learning machine; CNN: convolutional neural network; RMSE: root mean square error; WELM: weighted extreme learning machine.

exceeds 15 °C, the aluminum reduction cell is “HOT” Label.

Comparison with related algorithms

To evaluate the performances of our proposed method, we compare several existing algorithms, such as SVM, ELM, and CNN. All the results are presented in Table 1. From the experiment results, we conclude that the accuracy rate is competent with other superheat identification methods, not only that the training time is shorter. It shows that our proposed method have better accuracy rate than the existing methods. The accuracy of CNN-WELM has up to 0.94. Compared with SVM, our proposed method has a better accuracy and less training time, which is due to the reason that SVM is a linear classifier and CNN-WELM utilizes the nonlinear wavelet kernel function. A closed solution of the hidden weights improved the computation efficiency.

Figure 6 concludes some intuitive results with the different hidden nodes in these methods. Our proposed method has the lowest error rate with the increasing hidden layer.

Figures 5, 7, and 8 give some related results of CNN with some combination of other activation functions, such as CNN + ELM (Sig), CNN + ELM (Tanh), and CNN + RBF. In the CNN method with a Gaussian RBF, for a candidate set of 50, 100, 200, and 400, the testing accuracy is 60%, 65%, 69%, and 77%, respectively. The Sigmoid kernel with a CNN and ELM method has a competitive performance with the

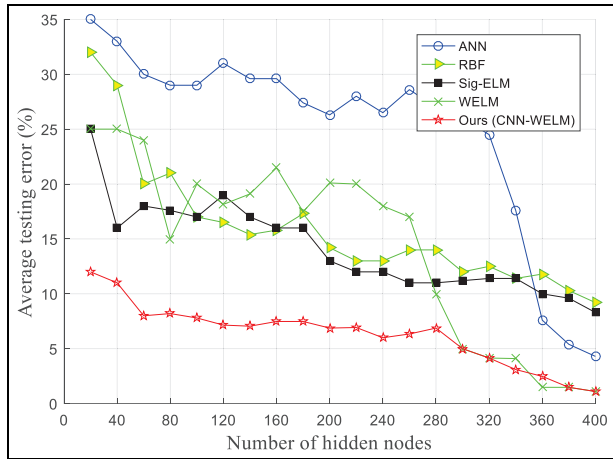


Figure 6. The performance comparison with the different algorithms. The average testing error denotes the number of misclassification with the 100 testing samples.

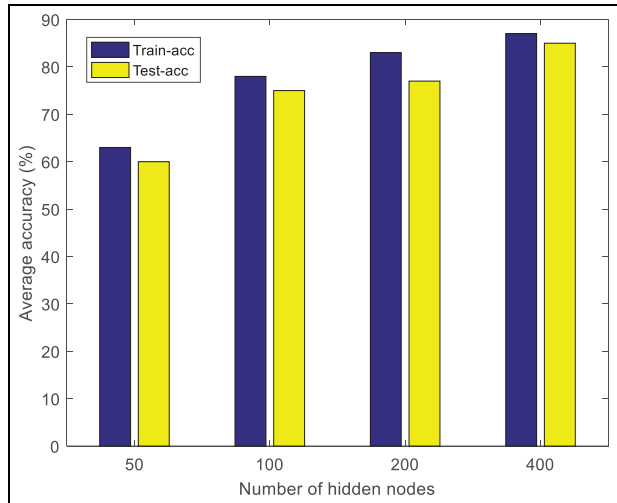


Figure 8. The accuracy of CNN-ELM (Tanh activation) with the different hidden units.

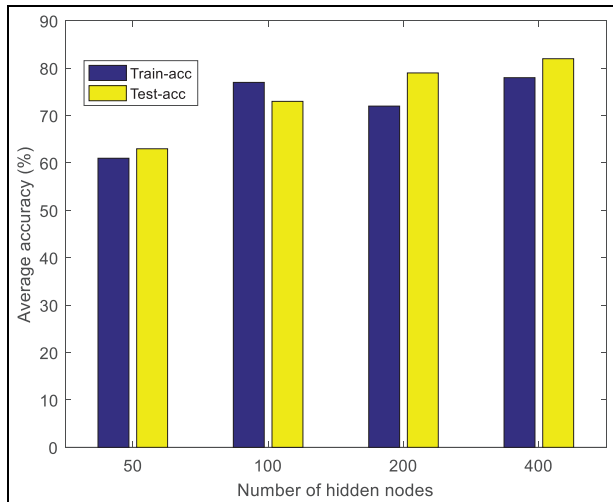


Figure 7. The accuracy of CNN-ELM (Sigmoid activation) with the different hidden units.

kernel of Tanh, and the corresponding testing accuracy rate is 63%, 73%, 79%, and 82%. These pictures demonstrate that with the increase in the hidden layers, the training and testing accuracy increased. The Tanh

activation used in CNN with ELM classifier has better accuracy, which means that the Tanh activation has a better feature ability. However, to make a comparison with the wavelet activation function, the superior performance of the proposed CNN-WELM has been obtained.

Conclusion

To solve the difficulty of the traditional manual determination of electrolytic cell conditions, we proposed a novel fire hole image recognition method based on CNN and wavelet ELM for aluminum electrolyzer. Furthermore, to improve the flexibility of models, the wavelet activation function is used in this article. The scale and shift factor make the proposed model having a better location property and flexibility. Compared with some of the existing state-of-the-art such as ELM, SVM, and CNN, the results show that the ELM has the advantages of less training time, strong generalization ability, and good robustness. The method proposed is not only limited to the prediction of aluminum electrolysis plant performance but also applies to the prediction of other industries. The model based on ELM has strong practical value and is worth

Algorithm 2. The proposed WELM Algorithm

Input: The original training flame hole images $\{(x_i, t_i)\}_{i=1}^N$, Morlet wavelet activation function $J(x)$, the learning rate of CNN; the hidden nodes N_{elm} ; the shift and transmission factors a and b .

Output: The hidden parameters of CNN; the output weight matrix β for ELM; the mapping function of f .

Step 1: Pre-processing the flame hole samples according to equation. (Normalization, standardization);

Step 2: Divide the samples into the training set, validation set, and testing set;

Step 3: Train the CNN neural network with the training dataset. The last hidden features are used to the input of the WELM.

Step 4: Initiate an ELM framework of n_{elm} hidden neurons with random weights and biases;

Step 5: Calculate the output of the hidden neurons J

Step 6: If $n \geq h_N$, compute the output weights using equation (7); Else, compute the output weights using equation (8);

step 7: Compute the projection function of ELM.

return The output weight matrix β ;

promoting and extending other domains. Our future work will be focusing on the multiple-resolution analysis of the industrial aluminum signals. Local time frequency will be analyzed and discussed with wavelet kernel transform, and these works will be reported by our future literature.

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